

Quantum_Tele

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WEEK-02: QUANTUM TELEPORTATION (DAY-08): RAJIV KUMAR YADAV

```
[1]: import qiskit

from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

import numpy as np

print("Qiskit version:", qiskit.__version__)

# =====
# QUANTUM TELEPORTATION
#
# Qubit 0 = Alice's unknown quantum state
# Qubit 1 = Alice's entangled qubit
# Qubit 2 = Bob's qubit
# =====

qc = QuantumCircuit(3, 2)

# =====
# STEP 1: Prepare Alice's unknown state
#
# Here we prepare:
#
#  $|> = \cos(\theta/2)|0> + \sin(\theta/2)|1>$ 
#
# Change theta to represent different states.
# =====

theta = np.pi / 3

qc.ry(theta, 0)
```

```

qc.barrier()

# =====
# STEP 2: Create Bell pair between Alice and Bob
#
# Alice's qubit = qubit 1
# Bob's qubit   = qubit 2
# =====

qc.h(1)

qc.cx(1, 2)

qc.barrier()

# =====
# STEP 3: Alice performs Bell-state measurement
# =====

qc.cx(0, 1)

qc.h(0)

qc.barrier()

# =====
# STEP 4: Measure Alice's two qubits
# =====

qc.measure(0, 0)
qc.measure(1, 1)

qc.barrier()

# =====
# STEP 5: Bob's correction
#
# If Alice's measurement results are:
#
# q1 = 1 → apply X
# q0 = 1 → apply Z
#
# =====

```

```

with qc.if_test(
    (qc.clbits[1], 1)
):
    qc.x(2)

with qc.if_test(
    (qc.clbits[0], 1)
):
    qc.z(2)

# =====
# Display circuit
# =====

print("\nQuantum Teleportation Circuit:\n")

print(
    qc.draw(
        output="text"
    )
)

# =====
# SIMULATION
# =====

simulator = AerSimulator()

compiled = transpile(
    qc,
    simulator
)

result = simulator.run(
    compiled,
    shots=1024
).result()

print("\nSimulation completed.")

print("\nMeasurement results:")

print(

```

```
result.get_counts()
)
```

Qiskit version: 2.4.1

Quantum Teleportation Circuit:

```
q_0:  Ry( /3)          H  M          »
                                     »
q_1:          H      X      M      »
                                     »
q_2:          X          If-0  X  End-0 »
                                     »
c: 2/          c_1=0x1          »
                                0 1          »
«
«q_0:
«
«q_1:
«
«q_2:  If-0      Z      End-0
«
«
«c: 2/  c_0=0x1
«
```

Simulation completed.

Measurement results:

```
{'11': 251, '01': 247, '10': 261, '00': 265}
```

[]: